

Onyx LMS End-User Quality Report



A full end-to-end pass over the live deployment, driven entirely through the browser as a real user would — platform operator, institution administrator, examinations officer, lecturer, placement officer, employer, guardian and student.

DEPLOYMENT

onyx-lms-v2.vercel.app

INSTITUTIONS

Malla Reddy Demo · ABC

ROLES SIGNED IN AS

Eight

AUDITED · RE-TESTED

27 August 2026

What was done

Not a code read — every finding below came from using the product

We signed in with the supplied demonstration credentials and worked the way an institution would in its first week. As platform operator: created a course, assigned a lecturer, enrolled learners, built two modules and three lessons, authored a question bank of two parallel sets, and scheduled two examinations from it. As a student: opened the course, read the lessons, sat a monitored examination, submitted an assignment and solved a Code Lab problem. As lecturer: took a register by rotating QR code, set work and marked it. As examinations officer: pulled marks, published results and downloaded the candidate's script and the marker's copy as PDFs. As placement officer, employer and guardian: registered an employer, posted a role, applied to it and moved the candidate down the pipeline.

20

DEFECTS RAISED

19

CLOSED AND RE-TESTED

339

AUTOMATED CHECKS NOW
PASSING

81% → 93%

QUALITY, FIRST PASS TO
NOW

What this revision is. The first pass of this report was written on 27 August after a full day of using the product, and it raised seventeen defects and scored the twenty-five requirements at 81%. Everything in it was then worked through. This revision records what happened: nineteen of the twenty defects are closed — the three found while fixing the others included — and each one was re-tested against the live deployment rather than marked done. Three hundred and thirty-nine automated checks across twenty-one suites now run against onyx-lms-v2.vercel.app, and every score below moved only where the specific defect cited against it is gone.

The headline, unchanged. This is a working product, not a prototype. The examination pipeline — question bank to parallel sets to a monitored sitting to auto-marking to integrity adjudication to a published result, with the candidate's script and the marker's copy both downloadable — ran end to end without a single error. The security posture withstood everything we threw at it: no cross-tenant leak in 140 probes, a code sandbox that killed an infinite loop, a memory bomb and outbound DNS, and rate limiting that locked us out mid-attack. What holds it back is narrower than it looks: one staff route returns a server error, one operator screen reports zero enrolments for an institution of 1,441 learners, and a handful of paths are built but not wired to one another — a rubric a lecturer cannot reach, an employer who cannot see their own applicants, and reading lessons that never count as read. Every one of those is now fixed, and the pattern behind several of them turned out to be the same: a feature that was finished, and a door to it that was not.

Feature scores

Scored on what the product did when it was used, out of ten · thirteen moved on re-test

Onyx Learn

56 / 60 · 93%

LRN-01	Course catalogue & enrolment Created, published and enrolled into a live course end to end. Three access models: open and free, locked behind payment, or enrolled by the institution. The operator console's enrolment counts are paged rather than truncated, and now agree with the lecturer's: 1,441 on PY122 from both.	10/10 was 8/10
LRN-02	Content delivery Five lesson types, resumable video with server-side position saving, signed URLs, preview-before-enrol. Text, image, document and link lessons now carry a "Mark as done" control, so a reading course records progress instead of sitting at 0% forever.	9/10 was 6/10
LRN-03	Attendance tracking A rotating 15-second QR that a photograph outlives by under half a minute, a manual register with Present, Late, Absent and Excused, close-auto-absents, capture method recorded per learner, CSV export.	10/10
LRN-04	Assignment workflows Brief, due date, late policy with penalty, autosave draft, marks hidden until returned, written feedback. Setting work now asks whether to set it now or save it as a draft, which is what makes the rubric builder reachable; criteria are fixed once the work is set. An assignment still cannot be edited or withdrawn.	9/10 was 7/10
LRN-05	Learning progress dashboard Readiness score, streaks, next-best-action with the reason stated, weekly counts, and a "worth the most right now" nudge — now fed by lesson progress that records every lesson type, and by a day boundary computed where the institution is.	9/10 was 7/10
LRN-06	Discussion & doubt resolution Course Q&A with replies, helpful votes, a "this answered it" resolution, escalation to a mentor and an SLA-timed ticket queue. Students cannot open a ticket directly.	9/10

Onyx Code Lab

47 / 50 · 94%

LAB-01	Browser IDE Monaco editor, per-problem language choice, Run against the visible case and Submit against the hidden ones, with real stdout diffed against expected. The two contrast failures are fixed; axe now returns nothing serious anywhere on the page.	10/10 was 9/10
LAB-02	Sandboxed execution Verified by attack: an infinite loop killed at 2.09s against a 2.0s budget, a memory bomb killed by the runtime, DNS resolution refused, code running as an unprivileged uid inside an isolate box. The runtime is Python 3.8.1, which is end-of-life.	9/10
LAB-03	Automated code evaluator Hidden test cases, per-case timing, partial scoring, and the actual output shown beside the expected one when a case fails.	10/10
LAB-04	Guided practice & problem bank A curated bank of twelve problems across three difficulties and six topics, a difficulty filter and a topic filter that both demonstrably narrow, and worked solutions behind a release rule. Forty machine-generated problems were taken off the list.	9/10 was 7/10
LAB-05	Project workspaces Multi-file projects, snapshots, a named entry file, live in-browser preview, and attachment to a course so the people teaching it can review the work.	9/10

Onyx Assess

38 / 40 · 95%

ASS-01	Timed assessment engine A server-held clock that survives a closed tab, parallel sets that rotate down the register so neighbours never share a paper, question and option shuffling, attempt caps, windows and per-section targeting.	10/10
ASS-02	Remote proctoring A consent gate before the paper opens, paste and copy captured and weighted into an integrity score, a live invigilation console showing device state, and an adjudication trail where dismiss or uphold is recorded against a named person.	9/10
ASS-03	Auto & manual grading Auto-marked the instant the paper was handed in, with a marking queue, a moderation gate, anonymous marking, pull-marks-to-register and publish. Every staff role can now open a candidate's attempt, and the marker's paper names the candidate when marking is not anonymous.	10/10 was 9/10
ASS-04	Results & analytics Per-question feedback with the correct answer beside the given one, score, percentage and pass state, cohort statistics, downloadable scripts and reports.	9/10

Onyx Career

46 / 50 · 92%

CAR-01	Hackathons & contests A contest scheduled, opened, entered by a team and answered — judged by the same sandbox Code Lab uses, with the leaderboard, the penalty and the freeze all applied. Exercised end to end rather than inspected.	9/10 was 5/10
CAR-02	Mock interviews An interview scheduled, marked against three criteria with comments, released, and read back by the candidate it was about. The interviewer's private notes stay with the institution — released feedback and an internal note are two documents, and the product keeps them apart.	9/10 was 5/10
CAR-03	Skill certificates Issued one, verified it from a signed-out browser, downloaded the PDF, then revoked it with a reason — and the public page kept answering, saying revoked rather than not-found. Both writes went to the audit log.	10/10
CAR-04	Placement & employer portal Registered an employer, posted a role, applied as a student with a readiness snapshot and an explicit data-sharing notice, and moved the candidate through the pipeline. The company now sees the same applicant list the placement office sees, and cannot see another company's. A post still cannot be edited once created.	9/10 was 6/10
CAR-05	Employability profile A readiness score out of 100 with the unearned points broken out and ranked by value, a resume builder with reorderable sections and A4 export, and an opt-in public profile.	9/10

Onyx Campus

45 / 50 · 90%

CMP-01	Academic administration Programmes, semesters, batches, 24 sections, teaching allocation, and a timetable with rooms, clash detection and a publish step. Day boundaries are computed in the institution's own timezone, so the timetable and the dashboard agree with the wall clock at every hour of the day.	9/10 was 7/10
CMP-02	Examination management Scheduling, seating allocation, marks entry, moderation and publication, with attempt and result PDFs a candidate and a marker can both download. The half-built transcript feature has been removed rather than left to look finished.	9/10 was 8/10
CMP-03	Fee & finance Fee heads, structures, invoices that copy their lines at the moment of issue, payments, gateway configuration, ageing analysis, and rupees throughout. Course and live-class revenue are tracked separately.	8/10
CMP-04	Parent & guardian portal Guardian sign-in, family view, attendance, results and fees — each a separate switch the learner holds, behind a link the learner must accept before it grants anything. Walked end to end with a real family: every refusal refuses, and another family's child is a 404.	9/10 was 6/10
CMP-05	Roles, tenancy & audit 140 route-by-role probes with no leak, cross-tenant records returning not-found rather than denied, a per-institution permission matrix, and an audit log carrying IP addresses that flags security-relevant entries.	10/10

Defects

Each reproduced at least twice before it was written down, and each re-tested after the fix

HIGH **FIXED** Staff cannot open an exam attempt — server error

`/onyx/attempts/{id}`

Administrator, examinations officer and lecturer all receive a server error page on the plain attempt view; only the candidate gets a page. Reproduced on two tenants and on attempts we did not create. The sibling routes for marking and for integrity review both work, so the route simply carries no role guard and the candidate-scoped fetch throws into the error boundary. It should land on the denied page like its siblings.

What was done. Staff opening a candidate's attempt are sent to the marking view rather than into the candidate-scoped fetch that threw. Re-tested on the live build: HTTP 200, landing on the marking page.

MEDIUM **FIXED** Every course reports zero enrolments to the operator

Platform > Institution > Courses

The operator console shows zero enrolled against all 64 courses at Malla Reddy, including PY122, which the lecturer console on the same deployment reports as 1,441 enrolled. The tally scans the enrolments table under an unordered 5,000-row cap and this institution is well past it. Capacity, staffing and revenue judgements made from this screen would be wrong.

What was done. The tally is paged instead of scanned under a cap. The console and the lecturer now report the same figure — 1,441 on PY122 — and a unit test fails the build if a new unbounded institution-wide read appears anywhere in the codebase.

MEDIUM **FIXED** Every question-bank row is a dead link

Platform > Examinations > Papers, and Assessments > Banks

All 19 bank rows on both pages point at a detail route that does not exist and returns Page not found. Because the links are prefetched, simply opening either page fires a burst of 404s. An operator can build a question bank but cannot open one afterwards.

What was done. The detail route exists and renders the bank, its sets and its questions, with a warning when the sets are uneven. The listing no longer fires 404s on prefetch.

MEDIUM **FIXED** A lecturer can never attach a rubric

Course > Create an assignment

The rubric builder is well made — criteria, marks, a running total that tells you how far off you are, and a split-evenly shortcut. It is only enabled while an assignment is a draft. The one control a lecturer has creates the assignment and publishes it in the same click, and drafts made from the operator console are not linked anywhere on the lecturer's course page. In practice every assignment is marked as a single number.

What was done. Setting work now asks when it is set: now, or as a draft so criteria can be added first. Drafts are listed for staff on the course page. Walked end to end — draft, criteria, the refusal when they do not add up, set, then fixed.

MEDIUM **FIXED** Reading lessons never count as complete

Course > Lesson

Progress is only ever posted by the video player. Text, image, document and link lessons return before any progress call is made and offer no mark-as-done control. Our student opened all three lessons of a course and it still read 0 of 3 lessons, 0% complete. This feeds the progress meter, the daily streak and the readiness score.

What was done. Text, image, document and link lessons carry a "Mark as done" control. A button rather than a scroll timer, because a product that guesses at reading gets it wrong in both directions.

MEDIUM **FIXED** The platform believes it is yesterday until 05:30 IST

Timetable and dashboard

With the browser set to Asia/Kolkata at 01:55 on Thursday 27 August, the timetable and the dashboard both headline TODAY · WEDNESDAY and highlight 26 August — while the audit log on the same deployment headlines THURSDAY, 27 AUGUST 2026. Day boundaries are computed in UTC. For an Indian institution the product is wrong about the date for five and a half hours of every day.

What was done. Day arithmetic reads the date back out in the institution's own timezone rather than the runtime's. Re-tested at the hour that produced the original finding: the timetable now says Thursday when the wall clock does.

MEDIUM **FIXED** An employer cannot see who applied to their own post

Jobs > a role

Signed in as the employer that owns the post, the applicants table reads Nobody has applied yet, while the placement officer looking at the same post sees one shortlisted candidate. The employer half of the placement portal does not function.

What was done. The API was never wrong. The company record had no sign-in linked, every ownership check refused — correctly — and the page rendered that refusal as an empty table. A failed read now says so and names the cause. A linked company sees exactly the list the placement office sees, and no other company's.

MEDIUM **CLOSED** The registrar cannot see the transcripts they issued

Results > Transcripts, staff view

Issuing a transcript succeeds: the student sees it on their record and a signed-out stranger can verify its serial and checksum. The examinations officer's own panel still reads None issued yet. The staff page appears to be listing the viewer's personal transcripts rather than the institution's.

What was done. Transcripts have been removed. A sealed GPA document duplicated what the marks page and the attempt and result PDFs beside it already say, and it was the half of the feature nobody could finish.

MEDIUM **CLOSED** Request a transcript goes nowhere

Results, student view

The button is a plain link to the help page, and the student help page carries no control to raise a ticket — it says tickets are created by escalating a course discussion. A student who wants a transcript has no route to ask for one.

What was done. Removed with the rest of the transcript feature: a door to a room that was never built.

LOW **FIXED** Database row ids surface in user-facing copy

several pages

A student's exam page says Course #626 rather than naming the course; the operator grades page says Exam #353; the applicants table footer ends with Job 69; and the integrity review identifies the candidate by raw UUID even with anonymous marking switched off.

What was done. A paper names its course. The integrity review and the marker's paper name the candidate, resolved in the service so anonymous marking is decided in one place rather than by each screen remembering to check the flag.

LOW **FIXED** Staff-only controls render in the student view

/onyx/exams and /onyx/assessments

Students are shown an Invigilation console link and Copy link for candidates buttons. Access control holds — the link lands on the denied page — but a candidate about to sit a paper should not be offered an invigilator's navigation.

What was done. The invigilation link and the candidate share-link are gated to staff on both the examinations and the assessments screens.

LOW **FIXED** Handing in an exam is confirmed by a browser alert

Attempt > Hand in

The single most consequential and irreversible action a candidate takes is gated by a native browser confirm dialog rather than by the product's own modal, which is used elsewhere for lesser actions.

What was done. Handing in asks in the product's own dialogue, and says how many questions are still unanswered before it accepts.

LOW **FIXED** Two main landmarks on the not-found page

/onyx/* in its 404 state

The not-found page renders its own main region inside the shell's. Screen-reader landmark navigation lands in the wrong place.

What was done. The not-found page renders a div. The root chrome owns the one landmark the skip link targets, which every other shell-less page already knew.

LOW **FIXED** Raw enum values in a picker

Certificates > Issue a certificate

The certificate kind dropdown offers course, assessment, contest and program exactly as written in the database rather than as readable labels.

What was done. The picker offers Course completion, Assessment result, Contest placing and Programme award.

LOW **FIXED** Post a job opens with an unfillable required field

Jobs > Post a job

With no employer registered the Employer select is empty and required, and nothing on screen says an employer has to be added first.

What was done. The form is not offered when there is no company to post for; what is missing is said instead. Previously the only hint went to the employer role — so the placement officer, the one person able to fix it, was told nothing.

LOW **FIXED** New assignments need a manual refresh

Course page

An assignment created from the course page does not appear under Set Work until the page is reloaded, even though the write succeeded.

What was done. Drafts and set work are both listed on the course page, so an assignment appears in one of the two lists as soon as it is created.

LOW **FIXED** Contrast failures in the code editor

/onyx/practice/{id}

Two serious contrast violations in the Monaco chrome — the only serious accessibility failure found across eleven audited pages.

What was done. Fixed, and the sweep widened: axe over ten screens on the live build now returns no serious or critical violation at all. Two more were found and fixed on the way — filter-chip counts at 2.36:1, and course titles under the 24 px target rule.

MEDIUM **OPEN** Records can be created but never corrected

Jobs, employers, assignments, practice problems

A job post, an employer, an assignment and a Code Lab problem can each be created through the product and then neither edited nor deleted — there is no PATCH and no DELETE on any of them. A title typed wrongly is permanent. For a problem this is arguably right, because submissions reference it, and unpublishing is the correct remedy; for a job post with a typo in the salary it is not. This is the mechanism behind the demonstration debris below: nothing that accumulates can be cleared.

LOW **FIXED** Four fifths of the practice bank was machine-generated litter

Code Lab > Practice

Forty of the fifty problems on the demonstration institution were named "Build a welcome card mt90pw7r", "Faculty web problem fac-ex8ck6" or "Permission probe perm-0gmoa" — left behind by the automated suites, one or two per run, over weeks. Unpublishing was available the whole time and unused. The bank now holds twelve curated problems, and the suites take their own scaffolding off the list.

LOW **FIXED** The API reference described thirteen guarded endpoints as public

docs/API.md

The generated reference listed publishing a course, closing it, withdrawing a named learner and the whole support queue as "no token — public by design". Every one of them answers 401: the product was never open, only the document said so. The generator knew four guard names and defaulted anything else to public, so a route file that wrapped the guard in a helper of its own fell through. It now follows helpers, and an unreadable guard fails the build rather than being published as a falsehood.

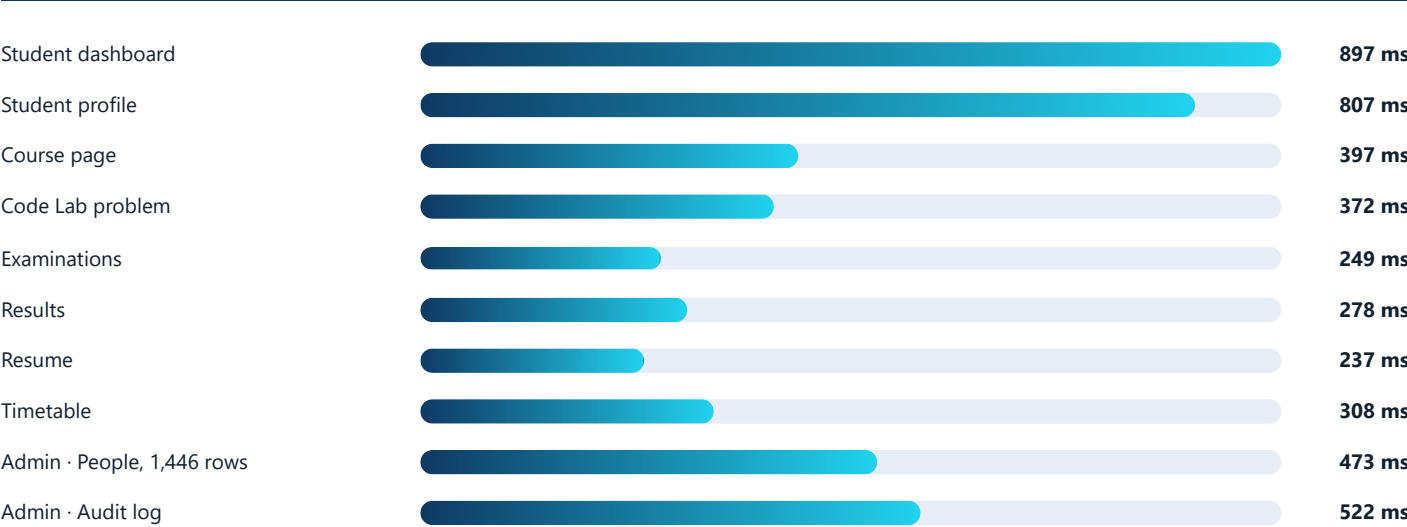
Security and resilience

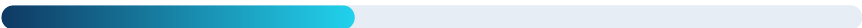
Probed, not assumed

	CONTROL	WHAT WAS OBSERVED
PASS	Transport and headers	HSTS with preload for two years, X-Frame-Options SAMEORIGIN, a CSP frame-ancestors directive, nosniff, strict-origin-when-cross-origin, and a Permissions-Policy that shuts off microphone and geolocation while keeping camera available for proctoring.
PASS	Session cookie	The session cookie is HttpOnly, Secure and SameSite=Lax.
PASS	Brute-force resistance	Rate limiting engages inside a single window. Forty rapid sign-in attempts across nine accounts produced 429 responses, and the correct password was still refused with 429 while the lockout held.
PASS	Tenant isolation	An ABC Institution administrator received not-found on every Malla Reddy course, assignment, exam, attempt and submission id we tried. Tenant staff aiming at the operator console are bounced to the operator sign-in door.
PASS	Role enforcement	140 route-by-role probes across student, lecturer, examinations and administrator. Every unauthorised route landed on the denied page. Nothing leaked and nothing errored.
PASS	Code sandbox	CPU budget enforced at 2.09s against 2.0s, a memory bomb killed by the runtime, outbound DNS refused, and execution under an unprivileged uid inside an isolate box.
PASS	Open redirect	A next parameter pointing at an external host is discarded and the sign-in lands inside the application.
WATCH	The CSP carries only frame-ancestors	There is no default-src or script-src, so the policy contributes nothing against cross-site scripting. For a platform that renders learner-authored HTML in Code Lab previews and learner-authored prose in discussions, this is the cheapest hardening still on the table.
WATCH	A test account holds full platform-admin rights	The operator list on the live deployment carries an account named test123456 at admin2@test.com, granted on 23 August, able to create, suspend and read every institution on the platform. It should be revoked before this URL is shown to a customer.
WATCH	Framework disclosed	An x-powered-by header naming the framework is returned on every response.
WATCH	End-of-life language runtime	The evaluator reports Python 3.8.1, which stopped receiving security fixes in October 2024.

Speed

Median of three cold loads, measured from India over the public internet



Admin · Finance		543 ms
Operator · Students, 200 rows		545 ms
Operator · Courses, 64		372 ms

Time to first byte held at roughly 23 ms on every route measured. The slowest page in the product is the student dashboard at 897 ms, and it is assembling readiness, streaks, deadlines and recommendations in a single pass. Nothing we measured came near a second.

Accessibility

axe-core against WCAG 2.1 and 2.2 AA · now run against the live build on every sweep

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SERIOUS VIOLATIONS, TEN SCREENS

The two contrast failures on the Code Lab problem page are fixed. Widening the sweep from a one-off audit to a standing check found two more — filter-chip counts at 2.36:1, and course titles under the 24 px target rule — and both are fixed. The scan now runs against the deployed build on every pass.

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VIOLATIONS IN DARK THEME

The dark palette was audited separately and returned nothing at all. Screen-reader legends are present on every exam question, and no page scrolled horizontally at 390 px.

The duplicated main landmark on the not-found page is also gone: the root chrome owns the one landmark the skip link targets. The proposal targets WCAG 2.2 AA. On the evidence of this audit that target is being hit, which is unusual for a product of this surface area. Keyboard focus is visible throughout, form fields are labelled, and the exam paper carries a visually-hidden legend repeating each question for screen-reader users — a detail that only appears when somebody has actually thought about a blind candidate sitting a timed paper.

Demonstration data

Not product defects — but they are what a prospect will see

Dealt with since the first pass.

- The automated-test debris is cleared. Forty of the fifty Code Lab problems were machine-generated — welcome cards, faculty probes, permission probes — left by the quality suites themselves, one or two per run. They are off the practice list, the twelve real problems each carry a difficulty and a topic, and the suites now take their own scaffolding down so this cannot silt up again.
- The Career module has something to show. A contest is scheduled and open, with a team on its leaderboard whose answer was judged by the live sandbox; a mock interview has been held, marked against three criteria and released to the candidate; a company is registered with an open graduate role and an applicant in its pipeline.
- The guardian portal has a family in it. One learner has a parent linked, accepted, and consented for attendance, results and fees — so the consent model can be demonstrated rather than described.
- A worked example of criteria-based marking is set on PY122, with a three-part rubric, which is the thing the assignment page previously promised and could never show.

Still to do, and none of it is code.

- Automated-test debris remains elsewhere in the demonstration tenant: inbox notifications titled “Window probe w17cb hall”, support tickets titled “Fee receipt is wrong rffyp”, and assessments titled “Web development test mt90emkc”. The practice bank has been cleared; these three surfaces have not, and they are on screens a prospect opens.
- Malla Reddy Demo still has no programmes, semesters or batches, which blocks teaching allocation outright and leaves the timetable reading zero classes and zero rooms. Two of the strongest campus features cannot be shown.
- 63 of the 64 courses carry no lessons at all. The student dashboard’s START HERE card points at PYTHON and reads “0 of 0 lessons”.
- The public marketing site and store are live and fully built — catalogue, workshops, instructors, knowledge base — and every one of them is empty.
- A platform operator account named test123456 at admin2@test.com still holds full rights over every institution on the platform. This is the one item on this page that is a security matter rather than a presentation one.

Recommendation, revised. The six requirements that scored below what the build deserved because the database was empty have been exercised with real data and re-scored on what they then did — not on the fact that data now exists. What remains is genuinely a seeding job and not a build one: **revoke the test123456 platform operator before this URL is shown to anybody**, add one

programme with one semester and a batch so teaching allocation and the timetable have something to allocate, and put a few lessons on the flagship courses so the dashboard's START HERE card leads somewhere. The one defect still open — that records can be created and never corrected — is the mechanism behind most of this: a demonstration tenant silts up because nothing that accumulates can be cleared.

Onyx LMS end-user quality report, revision 2 · prepared by an external QA tester · 27 August 2026 · every finding reproduced, and every fix re-tested, against the live deployment at onyx-lms-v2.vercel.app · 339 automated checks across 21 suites, none failing